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Course: Python 100 Assignment 05_ Model 05

Assignment 05 - Python Programming Basics

Module 05 - Collections of data

Introduction

Create a Python program that demonstrates using constants, variables, and print statements to display a message about a student's registration for a Python course. This program is very similar to Assignment04, but **It adds the use of data processing using dictionaries and exception handling.**

File Name:

- The file is named Assignment05.py

Script Header:

- The script header includes this text and has been updated with your name and the current date.

Constants:

- **The constant MENU: str is set to the value:**

```
---- Course Registration Program ----
```

```
Select from the following menu:
```

```
1. Register a Student for a Course
```

```
2. Show current data
```

```
3. Save data to a file
```

```
4. Exit the program
```

```
-----
```

- **The constant FILE_NAME: str is set to the value "Enrollments.csv"**
- **The constant values do not change throughout the program.**

Variables:

- **student_first_name: str is set to empty string.**
- **student_last_name: str is set to empty string.**
- **course_name: str is set to empty string.**

- **file is set to None**
- **menu_choice: str is set to empty string.**
- **student_data: dict is set to an empty dictionary (This is changed from a list using in assignment04)**
- **students: list: list is set to and empty list**

Input / Output:

- On menu choice 1, the program prompts the user to enter the student's first name and last name, followed by the course name, using the input() function and stores the inputs in the respective variables.
- Data collected for menu choice 1 is added to a dictionary named student_data. Next, student_data is added to the **students** two-dimensional list of dictionaries rows.
- On menu choice 2, the presents a string by formatting the collected data using the print() function.
- On menu choice 2, the program uses the print() function to show a string of comma-separated values for each row collected in the **students** variable.

Summary: Please refer to the screenshots

1, run the script to display the menu; choose 1 to add new data

```
C:\Users\Frank\Downloads\Module03\Module03\venv\Scripts\python.exe "C:\Users\Frank\Downloads\Module03\homework_result\Assignment05_FrankLi.py"
The enrollment file contains invalid data.

---- Course Registration Program ----

Select from the following menu:

1. Register a Student for a Course
2. Show current data
3. Save data to a file
4. Exit the program
-----

Enter your menu choice: 1
Enter student's first name: Frank Li
Enter student's last name: Li
Enter course name: Python 002

Student successfully registered.
```

2, choose 2 to show the input values

```

---- Course Registration Program ----

Select from the following menu:

1. Register a Student for a Course
2. Show current data
3. Save data to a file
4. Exit the program

-----

Enter your menu choice: 2

Current Enrollment Data
-----
Frank Li, Li, Python 002

```

3, choose 3 to save the input to Enrollments.json; and check json file

```

-----

Enter your menu choice: 3

Data saved successfully.

Current file contents:
Frank Li, Li, Python 002

```

```

Assignment05_FrankLi.py  Enrollments.json x
1
2  {
3    "FirstName": "Frank Li",
4    "LastName": "Li",
5    "CourseName": "Python 002"
6  }
7

```

4, repeat above steps to add more values

```

Enter your menu choice: 1
Enter student's first name: Frank
Enter student's last name: Li
Enter course name: C# 002

Student successfully registered.

```

```

Enter your menu choice: 2

Current Enrollment Data
-----
Frank Li, Li, Python 002
Frank, Li, C# 002

```

5, choose 2 to display all input values

```
Enter your menu choice: 2

Current Enrollment Data
-----
Frank Li, Li, Python 002
Frank, Li, C# 002
```

6, choose 3 to save the values; then check values in json file

```
Enter your menu choice: 3

Data saved successfully.

Current file contents:
Frank Li, Li, Python 002
Frank, Li, C# 002
```

```
{
  {
    "FirstName": "Frank Li",
    "LastName": "Li",
    "CourseName": "Python 002"
  },
  {
    "FirstName": "Frank",
    "LastName": "Li",
    "CourseName": "C# 002"
  }
}
```

7, choose 4 to quit the menus

```
Enter your menu choice: 4

Thank you for using the Course Registration Program.
Program ended.

Process finished with exit code 0
```

8, error handling on blank inputs

```
Enter your menu choice: 1
Enter student's first name:
First name cannot be blank.
Enter student's first name:
First name cannot be blank.
Enter student's first name:
First name cannot be blank.
Enter student's first name: Frank
Enter student's last name:
Last name cannot be blank.
Enter student's last name: Li
Enter course name:
Course name cannot be blank.
```

Result: the script can use the constant variable; and options from menu; and successfully add/save the new inputs by using functions (open(), write(), close()), print the correct output; and save the inputs to json file by following the chosen value.

Error handling

- The program provides structured error handling when the file is read into the list of dictionary rows.
- The program provides structured error handling when the user enters a first name.
- The program provides structured error handling when the user enters a last name.
- The program provides structured error handling when the dictionary rows are written to the file.